

Skill Week 9

Aim

To learn how to **list active background jobs**, display their status, identify job IDs and process IDs, and monitor the current state of jobs in Linux.

```
navaneeth@Navaneeth: ~/ski x + v
navaneeth@Navaneeth:~$ mkdir -p skillweek9
navaneeth@Navaneeth:~$ cd skillweek9
navaneeth@Navaneeth:~/skillweek9$ sleep 60 &
sleep 90 &
[1] 482
[2] 488
navaneeth@Navaneeth:~/skillweek9$ jobs
[1]-  Running                  sleep 60 &
[2]+  Running                  sleep 90 &
navaneeth@Navaneeth:~/skillweek9$ jobs -l
[1]-  482 Running                sleep 60 &
[2]+  488 Running                sleep 90 &
navaneeth@Navaneeth:~/skillweek9$ ps -o pid,ppid,pgid,stat,cmd
  PID   PPID   PGID STAT  CMD
   325    324    325  Ss    -bash
   482    325    482  S      sleep 60
   488    325    488  S      sleep 90
   516    325    516  R+     ps -o pid,ppid,pgid,stat,cmd
navaneeth@Navaneeth:~/skillweek9$ jobs -s
navaneeth@Navaneeth:~/skillweek9$ jobs -r
[1]-  Running                  sleep 60 &
[2]+  Running                  sleep 90 &
navaneeth@Navaneeth:~/skillweek9$ sleep 5
navaneeth@Navaneeth:~/skillweek9$ jobs
[1]-  Running                  sleep 60 &
[2]+  Running                  sleep 90 &
navaneeth@Navaneeth:~/skillweek9$ kill %1
[1]-  Terminated              sleep 60
navaneeth@Navaneeth:~/skillweek9$ kill %2
[2]+  Terminated              sleep 90
navaneeth@Navaneeth:~/skillweek9$ jobs
navaneeth@Navaneeth:~/skillweek9$ |
```

Observation

The jobs command lists background jobs and their states. The jobs -l command provides additional information such as the process ID. The ps command displays detailed process information.

Result

Active background jobs were successfully **listed, identified, monitored, and terminated** using Linux job-control commands.

Aim

To learn how to identify a background job, bring it to the foreground, transfer terminal control, resume a stopped process, and wait for the process to complete.

```
navaneeth@Navaneeth:~/skillweek9$ sleep 30 &
[1] 596
navaneeth@Navaneeth:~/skillweek9$ jobs -l
[1]+  596 Running                  sleep 30 &
navaneeth@Navaneeth:~/skillweek9$ fg %1
sleep 30
^C
navaneeth@Navaneeth:~/skillweek9$ jobs
navaneeth@Navaneeth:~/skillweek9$ sleep 20 &
[1] 636
navaneeth@Navaneeth:~/skillweek9$ jobs
[1]+  Running                    sleep 20 &
navaneeth@Navaneeth:~/skillweek9$ fg %1
sleep 20
navaneeth@Navaneeth:~/skillweek9$ jobs
navaneeth@Navaneeth:~/skillweek9$ |
```

Observation

The jobs command identifies active jobs, while fg transfers a background job to the foreground and gives it terminal control. Ctrl + C interrupts the foreground process. After a process completes, it is removed from the active job list.

Result

Background jobs were successfully **identified, transferred to the foreground, monitored, interrupted, and completed** using Linux job-control commands.

Aim

To understand how **Linux signals** are delivered to processes, how terminal signals such as SIGINT and SIGSTOP affect processes, and how to observe signal behavior

using process-

```
navaneeth@Navaneeth:~/skillweek9$ sleep 60 &
[1] 679
navaneeth@Navaneeth:~/skillweek9$ jobs -l
[1]+  679 Running                  sleep 60 &
navaneeth@Navaneeth:~/skillweek9$ ps -o pid,ppid,pgid,sid,stat,cmd
  PID   PPID   PGID   SID  STAT  CMD
   325    324    325    325  Ss    -bash
   679    325    679    325  S     sleep 60
   698    325    698    325  R+    ps -o pid,ppid,pgid,sid,stat,cmd
navaneeth@Navaneeth:~/skillweek9$ kill -STOP PID
-bash: kill: `PID': not a pid or valid job spec
navaneeth@Navaneeth:~/skillweek9$ kill -STOP 679

[1]+  Stopped                  sleep 60
navaneeth@Navaneeth:~/skillweek9$ ps -o pid,pgid,stat,cmd
  PID   PGID  STAT  CMD
   325    325  Ss    -bash
   679    679  T     sleep 60
   766    766  R+    ps -o pid,pgid,stat,cmd
navaneeth@Navaneeth:~/skillweek9$ kill -CONT 679
[1]+  Done                  sleep 60
navaneeth@Navaneeth:~/skillweek9$ jobs
navaneeth@Navaneeth:~/skillweek9$ kill 679
-bash: kill: (679) - No such process
navaneeth@Navaneeth:~/skillweek9$ jobs
navaneeth@Navaneeth:~/skillweek9$ |
```

Observation:

The system reports No such process because PID 679 no longer exists.

Result:

The process has already ended, so no further kill command is needed.

Aim

To learn how to **resume a stopped background job using SIGCONT**, check its status, and verify that the job continues execution successfully.

```

navaneeth@Navaneeth:~/skillweek9$ sleep 60 &
[1] 822
navaneeth@Navaneeth:~/skillweek9$ jobs -l
[1]+  822 Running                  sleep 60 &
navaneeth@Navaneeth:~/skillweek9$ kill -STOP 822

[1]+  Stopped                      sleep 60
navaneeth@Navaneeth:~/skillweek9$ jobs
[1]+  Stopped                      sleep 60
navaneeth@Navaneeth:~/skillweek9$ kill -CONT 822
navaneeth@Navaneeth:~/skillweek9$ jobs
[1]+  Running                     sleep 60 &
navaneeth@Navaneeth:~/skillweek9$ ps -o pid,pgid,stat,cmd
  PID   PGID  STAT  CMD
   325   325  Ss    -bash
   822   822  S     sleep 60
   891   891  R+    ps -o pid,pgid,stat,cmd
navaneeth@Navaneeth:~/skillweek9$ sleep 5
[1]+  Done                        sleep 60
navaneeth@Navaneeth:~/skillweek9$ jobs
navaneeth@Navaneeth:~/skillweek9$ jobs -l
navaneeth@Navaneeth:~/skillweek9$ |

```

Observation

The background process was stopped using SIGSTOP and successfully resumed using SIGCONT. The jobs command showed the change from **Stopped** to **Running**.

Result

The stopped background job was successfully **resumed, monitored, and verified** using SIGCONT and Linux job-control commands.